

BHV_CutRange, BHV_Trail, and BHV_Shadow

Contact-relative following and positioning behaviors.

Included MIT MOOS-IvP PDF sources:

- bhv_cutrange.pdf
- bhv_trail.pdf
- bhv_shadow.pdf

The CutRange Behavior

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1 The CutRange Behavior

The CutRange behavior will drive ownship to reduce the range between itself and another specified vehicle (nearly the opposite of the BHV_AvoidCollision behavior).

1.1 The Patience Parameter

The behavior reasons about candidate ownship maneuvers and will use a combination of immediate-rate-of-closure (IROC) and expected closest point of approach (CPA) distance, to evaluate candidate maneuvers. Using solely IROC is regarded as having zero patience, and using solely CPA is regarded as having maximal patience. The idea is conveyed in Figure 1 below. The former method may be preferred for a contact with frequent heading changes, while the latter may be preferred for contacts on a trajectory that rarely changes.

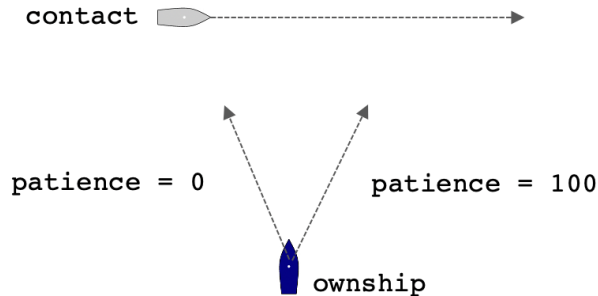


Figure 1: The *CutRange* behavior will either drive ownship directly toward the current contact position (zero patience setting) or drive ownship to be on the fastest intercept course based on the contact current position, heading and speed (maximal patience setting).

1.2 Automatic Priority Weight Variation Based on Range

The CutRange behavior may optionally be configured to automatically decrease its priority weight down to zero as it closes range to the contact. This can be used to ensure ownship does not collide with the contact or to transition the helm to a different mode of operation as it closes range to the contact. The idea is shown in Figure 2

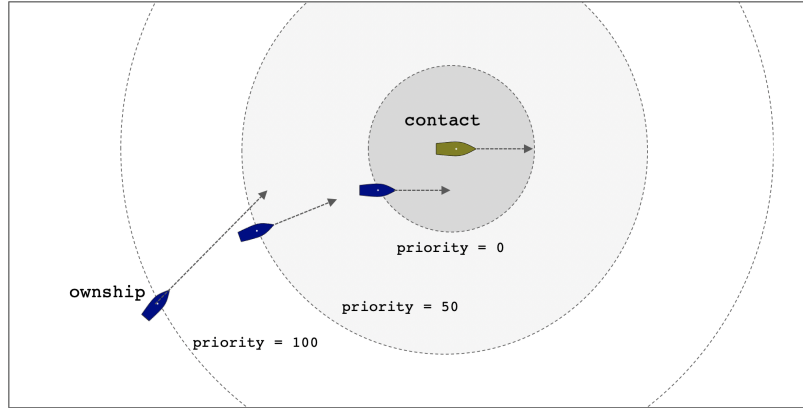


Figure 2: When ownship is at the outer range (`pwt_outer_range`), the priority is at its maximal value. When ownship is at the inner most range (`pwt_inner_range`), the priority weight drops to zero. The priority weight decreases linearly as the range is closed. If the overall configured priority weight for the behavior were set to 80 instead of 100, the outer weight would 80, and 40 at the intermediate range.

When the contact is at a range at or beyond `pwt_outer_dist`, the priority weight will be 100 percent of the behavior's configured priority weight. When the contact is at or closer than the range given by `pwt_inner_dist` the priority weight will be zero. At contact ranges between these two values the priority weight will linearly decrease in value. In the situation depicted in Figure 2, the vehicle transition to a dominant Shadow behavior as the range is closed. When the range decreases to `pwt_inner_dist`, the Shadow behavior is solely determining ownship heading and speed to match the contact's current heading and speed.

1.3 Initiating and Abandoning Pursuit

The `giveup_dist` parameter sets a contact range beyond which the behavior will assign, a zero priority weight, essentially giving up pursuit and becoming an inactive behavior. The parameter is typically set to be a bit higher than the `pwt_outer_dist` parameter, but any non-zero configured value will be respected. A `giveup_dist` of zero is the default, and disables this feature.

As ownship range to the contact increases and crosses the `giveup_dist` threshold, the behavior may be configured to post one or more messages, using the `giveupflag` parameter. Similarly, when ownship range to contact decreases below the giveup range, the behavior may be configured to post one or more messages, using the `pursueflag` parameter. For example:

```
pursueflag = PURSUIT=true  
giveupflag = PURSUIT=false
```

In the above case, the two flags could be used to transition the helm out of its present mission mode and back again based on the status of closing range to the contact.

1.4 Configuration Parameters

Listing 1.1: Configuration Parameters Common to All Behaviors.

<code>activeflag</code> :	A MOOS variable-value pair posted when the behavior is in the <i>active</i> state. [more] .
<code>condition</code> :	Specifies a condition that must be met for the behavior to be running. [more] .
<code>duration</code> :	Time in behavior will remain running before declaring completion. [more] .
<code>duration_idle_decay</code> :	When true, duration clock is running even when in the <i>idle</i> state. [more] .
<code>duration_reset</code> :	A variable-pair such as <code>MY_RESET=true</code> , that will trigger a duration reset. [more] .
<code>duration_status</code> :	The name of a MOOS variable to which the vehicle duration status is published. [more] .
<code>endflag</code> :	A MOOS variable-value pair posted when the behavior has completed. [more] .
<code>idleflag</code> :	A MOOS variable-value pair posted when the behavior is in the <i>idle</i> state. [more] .
<code>inactiveflag</code> :	A MOOS variable-value posted when the behavior is <i>not</i> in the <i>active</i> state. [more] .
<code>name</code> :	The (unique) name of the behavior. [more] .
<code>nostarve</code> :	Allows a behavior to assert a maximum staleness for a MOOS variable. [more] .
<code>perpetual</code> :	If true allows the behavior to run even after it has completed. [more] .
<code>post_mapping</code> :	Re-direct behavior output normally to one MOOS variable to another instead. [more] .
<code>priority</code> :	The priority weight of the behavior. [more] .
<code>pwt</code> :	Same as <code>priority</code> .
<code>runflag</code> :	A MOOS variable and a value posted when a behavior has met its conditions. [more] .
<code>spawnflag</code> :	A MOOS variable and a value posted when a behavior is spawned. [more] .
<code>spawnxflag</code> :	A MOOS variable and a value posted when a behavior is spawned. [more] .
<code>templating</code> :	Turns a behavior into a template for spawning behaviors dynamically. [more] .
<code>updates</code> :	A MOOS variable from which behavior parameter updates are read dynamically. [more] .

Listing 1.2: Configuration Parameters Common to Contact Behaviors.

Parameter	Description
<code>bearing_lines</code>	If true, a visual artifact will be produced for rendering the bearing line between ownship and the contact when the behavior is running. Not all behaviors implement this feature.
<code>contact</code>	Name or unique identifier of a contact to be avoided.
<code>decay</code>	Time interval during which extrapolated position slows to a halt.
<code>exit_on_filter_group</code>	If true, and if an exclusion filter is implemented for this contact behavior, an early exit of the behavior may be allowed when or if the group name changes and no longer satisfies the exclusion filter. The default is false.
<code>exit_on_filter_vtype</code>	If true, and if an exclusion filter is implemented for this contact behavior, an early exit of the behavior may be allowed when or if the vehicle type changes and no longer satisfies the exclusion filter. The default is false.
<code>exit_on_filter_region</code>	If true, and if an exclusion filter is implemented for this contact behavior, an early exit of the behavior may be allowed when or if the contact moves into a region that would no longer satisfy the exclusion filter. The default is false.
<code>extrapolate</code>	If true, contact position is extrapolated from last position and trajectory.
<code>ignore_group</code>	If specified, the contact group may not match the given ignore group. If multiple ignore groups are specified, the contact group must be different than all ignore groups. Introduced after Release 19.8.1. Section ??
<code>ignore_name</code>	If specified, the contact name may not match the given ignore name. If multiple ignore names are specified, the contact name must be different than all ignore names. Introduced after Release 19.8.1. Section ??
<code>ignore_region</code>	If specified, the contact group may be in the given ignore region. If multiple ignore regions are specified, the contact position must be external to all ignore regions. Introduced after Release 19.8.1. Section ??
<code>ignore_type</code>	If specified, the contact type may not match the given ignore type. If multiple ignore types are specified, the contact type must be different than all ignore types. Introduced after Release 19.8.1. Section ??
<code>match_group</code>	If specified, the contact group must match the given match group. If multiple match groups are specified, the contact group must match at least one match group. Introduced after Release 19.8.1. Section ??
<code>match_name</code>	If specified, the contact name must match the given match name. If multiple match names are specified, the contact name must match at least one. Introduced after Release 19.8.1. Section ??
<code>match_region</code>	If specified, the contact must reside in the given convex region. If multiple match regions are specified, the contact position must be in at least one match region. The multiple regions essentially can together support a non-convex regions. Introduced after Release 19.8.1. Section ??
<code>match_type</code>	If specified, the contact type must match the given match type. If multiple match types are specified, the contact type must match at least one match type. Introduced after Release 19.8.1. Section ??

<code>on_no_contact_ok</code>	If false, a helm error is posted if no contact information exists. Applicable in the more rare case that a contact behavior is statically configured for a named contact. The default is true.
<code>strict_ignore</code>	If true, and if one of the ignore exclusion filter components is enabled, then an exclusion filter will fail if the contact report is missing information related to the filter. For example if the contact group information is unknown. The default is true.
<code>time_on_leg</code>	The time on leg, in seconds, used for calculating closest point of approach.

Listing 1.3: Configuration Parameters for the CutRange Behavior.

Parameter	Description
<code>giveup_dist</code> :	Range to contact, in meters, outside which the behavior will give up (become inactive). Default is zero. Section 1.3.
<code>giveupflag</code> :	Optional MOOS variable-value pairs posted when ownship range to contact increases above <code>giveup_dist</code> . Section 1.3.
<code>patience</code> :	Linear scale choice between preferring heading directly to the contact (<code>patience=0</code>) or heading on the lowest closest point of approach (<code>patience=100</code>). Section 1.1.
<code>pursueflag</code> :	Optional MOOS variable-value pairs posted when ownship range to contact drops below <code>giveup_dist</code> . Section 1.3.
<code>pwt_outer_dist</code> :	Range to contact outside which the behavior has maximum priority. Section 1.2.
<code>pwt_inner_dist</code> :	Range to contact within which the behavior has zero priority. Section 1.2.

Listing 1.4: Example Configuration Block.

```

Behavior = BHV_CutRange
{
    // General Behavior Parameters
    // -----
    name          = cutrange_                // example
    pwt           = 200                      // example
    condition     = AVOID = true             // example
    updates       = CONTACT_INFO             // example
    endflag       = CONTACT_RESOLVED = ${CONTACT} // example
    templating    = spawn                   // example

    // General Contact Behavior Parameters
    // -----
    bearing_lines = white:0, green:0.65, yellow:0.8, red:1.0 // example

    contact = henry                // example
    decay   = 15,30                 // default (seconds)
    extrapolate = true             // default
    on_no_contact_ok = true        // default
    time_on_leg = 60               // default (seconds)

    // Parameters specific to this behavior
    // -----
    giveup_dist = 225              // Meters. Default is 0, disabled
    patience    = 45              // [0,100]. Default 0
    pwt_inner_dist = 50            // Meters. Default is 0.
    pwt_outer_dist = 200          // Meters. Default is 0.
    pursueflag = PURSUIT=true
    giveupflag  = PURSUIT=false
}

```

1.5 Variables Published

The below MOOS variables will be published by the behavior during normal operation, in addition to any configured flags. A variable published by any behavior may be suppressed or changed to a different variable name using the `post_mapping` configuration parameter described in Section ??.

- **VIEW_SEGLIST**: A bearing line between ownship and the contact if configured for rendering.
- Any variables involved in the **giveupflag** or **pursueflag** configurations, or other general behavior flags.

The Trail Behavior

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1 The Trail Behavior

This behavior will drive the vehicle to trail or follow another specified vehicle at a given relative position. A tool for "formation flying". The following parameters are defined for this behavior:

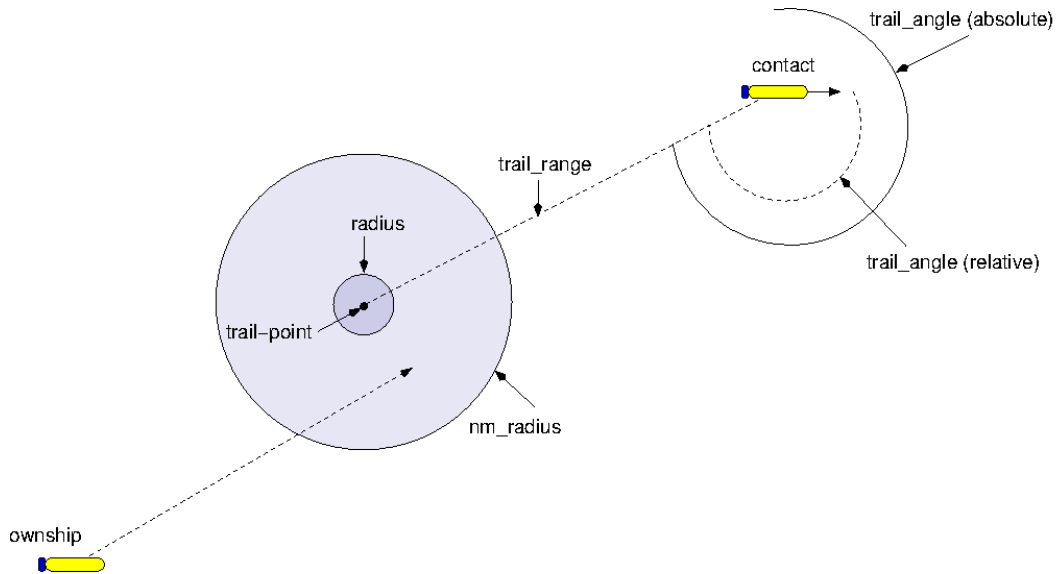


Figure 1: Interpolation of vehicle speed inside the radius set by `nm_radius` relative to the extrapolated trail position.

1.1 Configuration Parameters

Listing 1.1: Configuration Parameters Common to All Behaviors.

<code>activeflag</code> :	A MOOS variable-value pair posted when the behavior is in the <i>active</i> state. [more] .
<code>condition</code> :	Specifies a condition that must be met for the behavior to be running. [more] .
<code>duration</code> :	Time in behavior will remain running before declaring completion. [more] .
<code>duration_idle_decay</code> :	When true, duration clock is running even when in the <i>idle</i> state. [more] .
<code>duration_reset</code> :	A variable-pair such as <code>MY_RESET=true</code> , that will trigger a duration reset. [more] .
<code>duration_status</code> :	The name of a MOOS variable to which the vehicle duration status is published. [more] .
<code>endflag</code> :	A MOOS variable-value pair posted when the behavior has completed. [more] .
<code>idleflag</code> :	A MOOS variable-value pair posted when the behavior is in the <i>idle</i> state. [more] .
<code>inactiveflag</code> :	A MOOS variable-value posted when the behavior is <i>not</i> in the <i>active</i> state. [more] .
<code>name</code> :	The (unique) name of the behavior. [more] .
<code>nostarve</code> :	Allows a behavior to assert a maximum staleness for a MOOS variable. [more] .
<code>perpetual</code> :	If true allows the behavior to to run even after it has completed. [more] .
<code>post_mapping</code> :	Re-direct behavior output normally to one MOOS variable to another instead. [more] .
<code>priority</code> :	The priority weight of the behavior. [more] .
<code>pwt</code> :	Same as <code>priority</code> .
<code>runflag</code> :	A MOOS variable and a value posted when a behavior has met its conditions. [more] .
<code>spawnflag</code> :	A MOOS variable and a value posted when a behavior is spawned. [more] .
<code>spawnxflag</code> :	A MOOS variable and a value posted when a behavior is spawned. [more] .
<code>templating</code> :	Turns a behavior into a template for spawning behaviors dynamically. [more] .
<code>updates</code> :	A MOOS variable from which behavior parameter updates are read dynamically. [more] .

Listing 1.2: Configuration Parameters Common to Contact Behaviors.

Parameter	Description
<code>bearing_lines</code>	If true, a visual artifact will be produced for rendering the bearing line between ownship and the contact when the behavior is running. Not all behaviors implement this feature.
<code>contact</code>	Name or unique identifier of a contact to be avoided.
<code>decay</code>	Time interval during which extrapolated position slows to a halt.

<code>exit_on_filter_group</code>	If true, and if an exclusion filter is implemented for this contact behavior, an early exit of the behavior may be allowed when or if the group name changes and no longer satisfies the exclusion filter. The default is false.
<code>exit_on_filter_vtype</code>	If true, and if an exclusion filter is implemented for this contact behavior, an early exit of the behavior may be allowed when or if the vehicle type changes and no longer satisfies the exclusion filter. The default is false.
<code>exit_on_filter_region</code>	If true, and if an exclusion filter is implemented for this contact behavior, an early exit of the behavior may be allowed when or if the contact moves into a region that would no longer satisfy the exclusion filter. The default is false.
<code>extrapolate</code>	If true, contact position is extrapolated from last position and trajectory.
<code>ignore_group</code>	If specified, the contact group may not match the given ignore group. If multiple ignore groups are specified, the contact group must be different than all ignore groups. Introduced after Release 19.8.1. Section ??
<code>ignore_name</code>	If specified, the contact name may not match the given ignore name. If multiple ignore names are specified, the contact name must be different than all ignore names. Introduced after Release 19.8.1. Section ??
<code>ignore_region</code>	If specified, the contact group may be in the given ignore region. If multiple ignore regions are specified, the contact position must be external to all ignore regions. Introduced after Release 19.8.1. Section ??
<code>ignore_type</code>	If specified, the contact type may not match the given ignore type. If multiple ignore types are specified, the contact type must be different than all ignore types. Introduced after Release 19.8.1. Section ??
<code>match_group</code>	If specified, the contact group must match the given match group. If multiple match groups are specified, the contact group must match at least one match group. Introduced after Release 19.8.1. Section ??
<code>match_name</code>	If specified, the contact name must match the given match name. If multiple match names are specified, the contact name must match at least one. Introduced after Release 19.8.1. Section ??
<code>match_region</code>	If specified, the contact must reside in the given convex region. If multiple match regions are specified, the contact position must be in at least one match region. The multiple regions essentially can together support a non-convex regions. Introduced after Release 19.8.1. Section ??
<code>match_type</code>	If specified, the contact type must match the given match type. If multiple match types are specified, the contact type must match at least one match type. Introduced after Release 19.8.1. Section ??
<code>on_no_contact_ok</code>	If false, a helm error is posted if no contact information exists. Applicable in the more rare case that a contact behavior is statically configured for a named contact. The default is true.
<code>strict_ignore</code>	If true, and if one of the ignore exclusion filter components is enabled, then an exclusion filter will fail if the contact report is missing information related to the filter. For example if the contact group information is unknown. The default is true.

`time_on_leg` The time on leg, in seconds, used for calculating closest point of approach.

Listing 1.3: Configuration Parameters for the Trail Behavior.

Parameter	Description
<code>nm_radius</code> :	If in this range to contact and ahead of it, slow down. The default is 20 meters.
<code>no_alert_request</code> :	If true, no request will be made to the contact manager for contact alerts. The default is true.
<code>post_trail_dist_on_idle</code> :	If true, post <code>TRAIL_DISTANCE</code> even during the idle state. The default is true.
<code>pwt_outer_dist</code> :	Range to contact outside which the behavior has zero priority.
<code>radius</code> :	If outside this radius to the contact, head to <code>nm_radius</code> ahead of trail point. The default is 5 meters.
<code>trail_angle</code> :	Relative angle to the contact to set the trail-point. The default is 180.
<code>trail_angle_type</code> :	Either "relative" or "absolute" bearing/angle to the contact. The default is relative.
<code>trail_range</code> :	Relative distance to the contact to set the trail-point. The default is 50 meters.

Listing 1.4: Example Configuration Block.

```

Behavior = BHV_Trail
{
  // General Behavior Parameters
  // -----
  name          = trail_           // example
  pwt           = 100              // default
  condition     = TRAIL_ALLOWED = true // example
  updates       = TRAIL_INFO      // example
  templating    = spawn           // example

  // General Contact Behavior Parameters
  // -----
  contact = to-be-set             // example
  decay   = 15,30                 // default (seconds)
  extrapolate = true              // default
  on_no_contact_ok = true         // default
  time_on_leg = 60                // default (seconds)

  // Parameters specific to this behavior
  // -----
  nm_radius = 20                  // default (meters)
  no_alert_request = false        // default
  post_trail_dist_on_idle = true  // default
  pwt_outer_dist = 0              // default (meters)
  radius = 5                      // default (meters)
  trail_angle = 180               // default (degrees)
  trail_angle_type = relative     // default (or absolute)
  trail_range = 50                // default (meters)
}

```

1.2 Variables Published

The below MOOS variables will be published by the behavior during normal operation, in addition to any configured flags. A variable published by any behavior may be suppressed or changed to a different variable name using the `post_mapping` configuration parameter described in Section ??.

Listing 1.5: Variables Published by the Trail Behavior.

Variable	Description
<code>PURSUIT:</code>	1 if in behavior is active and producing an objective function, 0 otherwise.
<code>REGION:</code>	
<code>TRAIL_HEADING:</code>	
<code>TRAIL_SPEED:</code>	
<code>TRAIL_RANGE:</code>	
<code>MAX_RANGE:</code>	
<code>VIEW_POINT:</code>	
<code>TRAIL_DISTANCE:</code>	

1.3 Parameters

- `trail_range`: The range component of the relative position to the contact to trail.
- `trail_angle`: The relative angle of the relative position to the contact to trail. (180 is directly behind, 90 is a parallel track to the contacts starboard side, -90 is on the port side of the contact.)
- `trail_angle_type`: The trail angle may be set to either `relative` (the default), or `absolute`.
- `radius`: The distance (in meters) from the trail position that will result in the behavior “cutting range” to the trail position, and inside of which will result in the behavior “shadowing” the contact. The default is 5 meters.
- `nm_radius`: The distance in meters from the trail point within which the speed will be gradually change from the outer chase speed (max speed) and the speed of the contact, as illustrated in Figure 1. This parameter should typically be set to several times the value of `radius` to achieve smooth formation flying. Default is 20 meters.
- `max_range`: The distance (in meters) that the contact must be within for the behavior to be active and produce an objective function. The default is `max_range` value is zero meaning it will be active regardless of the distance to the contact.

The Shadow Behavior

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1 The Shadow Behavior

This behavior will drive the vehicle to match the trajectory of another specified vehicle. This behavior in conjunction with the BHV_CutRange behavior can produce a "track and trail" capability.

1.1 Configuration Parameters

Listing 1.1: Configuration Parameters Common to All Behaviors.

activeflag:	A MOOS variable-value pair posted when the behavior is in the <i>active</i> state. [more] .
condition:	Specifies a condition that must be met for the behavior to be running. [more] .
duration:	Time in behavior will remain running before declaring completion. [more] .
duration_idle_decay:	When true, duration clock is running even when in the <i>idle</i> state. [more] .
duration_reset:	A variable-pair such as <code>MY_RESET=true</code> , that will trigger a duration reset. [more] .
duration_status:	The name of a MOOS variable to which the vehicle duration status is published. [more] .
endflag:	A MOOS variable-value pair posted when the behavior has completed. [more] .
idleflag:	A MOOS variable-value pair posted when the behavior is in the <i>idle</i> state. [more] .
inactiveflag:	A MOOS variable-value posted when the behavior is <i>not</i> in the <i>active</i> state. [more] .
name:	The (unique) name of the behavior. [more] .
nostarve:	Allows a behavior to assert a maximum staleness for a MOOS variable. [more] .

<code>perpetual</code> :	If true allows the behavior to run even after it has completed. [more] .
<code>post_mapping</code> :	Re-direct behavior output normally to one MOOS variable to another instead. [more] .
<code>priority</code> :	The priority weight of the behavior. [more] .
<code>pwt</code> :	Same as <code>priority</code> .
<code>runflag</code> :	A MOOS variable and a value posted when a behavior has met its conditions. [more] .
<code>spawnflag</code> :	A MOOS variable and a value posted when a behavior is spawned. [more] .
<code>spawnxflag</code> :	A MOOS variable and a value posted when a behavior is spawned. [more] .
<code>templating</code> :	Turns a behavior into a template for spawning behaviors dynamically. [more] .
<code>updates</code> :	A MOOS variable from which behavior parameter updates are read dynamically. [more] .

Listing 1.2: Configuration Parameters Common to Contact Behaviors.

Parameter	Description
<code>bearing_lines</code>	If true, a visual artifact will be produced for rendering the bearing line between ownship and the contact when the behavior is running. Not all behaviors implement this feature.
<code>contact</code>	Name or unique identifier of a contact to be avoided.
<code>decay</code>	Time interval during which extrapolated position slows to a halt.
<code>exit_on_filter_group</code>	If true, and if an exclusion filter is implemented for this contact behavior, an early exit of the behavior may be allowed when or if the group name changes and no longer satisfies the exclusion filter. The default is false.
<code>exit_on_filter_vtype</code>	If true, and if an exclusion filter is implemented for this contact behavior, an early exit of the behavior may be allowed when or if the vehicle type changes and no longer satisfies the exclusion filter. The default is false.
<code>exit_on_filter_region</code>	If true, and if an exclusion filter is implemented for this contact behavior, an early exit of the behavior may be allowed when or if the contact moves into a region that would no longer satisfy the exclusion filter. The default is false.
<code>extrapolate</code>	If true, contact position is extrapolated from last position and trajectory.
<code>ignore_group</code>	If specified, the contact group may not match the given ignore group. If multiple ignore groups are specified, the contact group must be different than all ignore groups. Introduced after Release 19.8.1. Section ??
<code>ignore_name</code>	If specified, the contact name may not match the given ignore name. If multiple ignore names are specified, the contact name must be different than all ignore names. Introduced after Release 19.8.1. Section ??
<code>ignore_region</code>	If specified, the contact group may be in the given ignore region. If multiple ignore regions are specified, the contact position must be external to all ignore regions. Introduced after Release 19.8.1. Section ??

<code>ignore_type:</code>	If specified, the contact type may not match the given ignore type. If multiple ignore types are specified, the contact type must be different than all ignore types. Introduced after Release 19.8.1. Section ??
<code>match_group:</code>	If specified, the contact group must match the given match group. If multiple match groups are specified, the contact group must match at least one match group. Introduced after Release 19.8.1. Section ??
<code>match_name:</code>	If specified, the contact name must match the given match name. If multiple match names are specified, the contact name must match at least one. Introduced after Release 19.8.1. Section ??
<code>match_region:</code>	If specified, the contact must reside in the given convex region. If multiple match regions are specified, the contact position must be in at least one match region. The multiple regions essentially can together support a non-convex regions. Introduced after Release 19.8.1. Section ??
<code>match_type:</code>	If specified, the contact type must match the given match type. If multiple match types are specified, the contact type must match at least one match type. Introduced after Release 19.8.1. Section ??
<code>on_no_contact_ok</code>	If false, a helm error is posted if no contact information exists. Applicable in the more rare case that a contact behavior is statically configured for a named contact. The default is true.
<code>strict_ignore</code>	If true, and if one of the ignore exclusion filter components is enabled, then an exclusion filter will fail if the contact report is missing information related to the filter. For example if the contact group information is unknown. The default is true.
<code>time_on_leg</code>	The time on leg, in seconds, used for calculating closest point of approach.

Listing 1.3: Configuration Parameters for the Shadow Behavior.

Parameter	Description
<code>heading_peakwidth:</code>	Width of the peak, in degrees, in the produced ZAIC-style IvP function.
<code>heading_basewidth:</code>	Width of the base, in degrees, in the produced ZAIC-style IvP function.
<code>speed_peakwidth:</code>	Width of the peak, in m/sec, in the produced ZAIC-style function.
<code>speed_basewidth:</code>	Width of the base, in m/sec, in the produced ZAIC-style IvP function.

Listing 1.4: Example Configuration Block.

```

Behavior = BHV_CutRange
{
  // General Behavior Parameters
  // -----
  name          = shadow_henry          // example
  pwt           = 200                   // example
  condition     = SHADOW = true         // example
  updates       = CONTACT_INFO          // example

  // General Contact Behavior Parameters
  // -----
          contact = henry                // example
          decay   = 15,30                 // default (seconds)
          extrapolate = true              // default
  on_no_contact_ok = true                // default
          time_on_leg = 60                // default (seconds)

  // Parameters specific to this behavior
  // -----
          pwt_outer_dist = 0              // default (meters)
  heading_peakwidth = 20                 // default
  heading_basewidth = 160                // default
  speed_peakwidth   = 0.1                // default
  speed_basewidth   = 2.0                // default
}

```

- **pwt_outer_dist**: The distance (in meters) that the contact must be within for the behavior to be active and produce an objective function. The default is zero meaning it will be active regardless of the distance to the contact.
- **heading_peakwidth**: This behavior uses the ZAIC_PEAK tool from the IvP Toolbox for generating an objective function over heading and speed. This parameter sets the peakwidth parameter of the heading component.
- **heading_basewidth**: This behavior uses the ZAIC_PEAK tool from the IvP Toolbox for generating an objective function over heading and speed. This parameter sets the basewidth parameter of the heading component.
- **speed_peakwidth**: This behavior uses the ZAIC_PEAK tool from the IvP Toolbox for generating an objective function over heading and speed. This parameter sets the peakwidth parameter of the speed component.
- **speed_basewidth**: This behavior uses the ZAIC_PEAK tool from the IvP Toolbox for generating an objective function over heading and speed. This parameter sets the basewidth parameter of the speed component.

1.2 Variables Published

The below MOOS variables will be published by the behavior during normal operation, in addition to any configured flags. A variable published by any behavior may be suppressed or changed to a

different variable name using the `post_mapping` configuration parameter described in Section ??.

Listing 1.5: Variables Published by the Shadow Behavior.

Variable	Description
<code>SHADOW_CONTACT_X:</code>	description
<code>SHADOW_CONTACT_Y:</code>	description
<code>SHADOW_CONTACT_SPEED:</code>	description
<code>SHADOW_CONTACT_HEADING:</code>	description
<code>SHADOW_CONTACT_RELEVANCE:</code>	description